

HOOFSTUK 3

Differensiasie

3.1 Limiete

Beskou die volgende funksie: $y = \frac{5x}{x+1}$

Bereken die waardes van y vir $x=0$ tot $x=\infty$

x	y
0	0
1	2,50 00 00
2	3,33 33 33
3	3,75 00 00
4	4,00 00 00
5	4,16 66 67
10	4,54 54 55
50	4,90 19 61
100	4,95 04 95
1000	4,99 50 05
10^6	4,99 99 95

Wanneer $x \rightarrow \infty$ sal $y = \frac{\infty}{\infty}$ (ongedefineerd)

Die funksie kan herkryf word sodat

$$y = \frac{5x(\frac{1}{x})}{(x+1)(\frac{1}{x})} = \frac{5}{1 + \frac{1}{x}}$$

$$\therefore \text{wanneer } x \rightarrow \infty, \text{ sal } y = \frac{5}{1 + \frac{1}{\infty}} = \frac{5}{1+0} = 5$$

Die funksie $y = \frac{5x}{x+1}$ streef na die limietwaarde van

5,00 00 00 wanneer $x \rightarrow \infty$.

Dit word dan geskryf as :

$$\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} \frac{5x}{x+1} = \lim_{x \rightarrow \infty} \frac{5}{x + \frac{1}{x}} = \frac{5}{1 + \frac{1}{\infty}} = 5$$

CHAPTER 3

Differentiation

3.1 Limits

Consider the following function: $y = \frac{5x}{x+1}$

Determine the values of y for $x=0$ to $x=\infty$

x	y
0	0
1	2,50 00 00
2	3,33 33 33
3	3,75 00 00
4	4,00 00 00
5	4,16 66 67
10	4,54 54 55
50	4,90 19 61
100	4,95 04 95
1000	4,99 50 05
10^6	4,99 99 95

When $x \rightarrow \infty$ then $y = \frac{\infty}{\infty}$ (undefined)

The function can be rewritten in the form:

$$y = \frac{5x(\frac{1}{x})}{(x+1)(\frac{1}{x})} = \frac{5}{1 + \frac{1}{x}}$$

$$\therefore \text{when } x \rightarrow \infty, \text{ then } y = \frac{5}{1 + \frac{1}{\infty}} = \frac{5}{1+0} = 5$$

The function $y = \frac{5x}{x+1}$ strives to the limit value of

5,00 00 00 when $x \rightarrow \infty$.

It is then written as :

$$\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} \frac{5x}{x+1} = \lim_{x \rightarrow \infty} \frac{5}{x + \frac{1}{x}} = \frac{5}{1 + \frac{1}{\infty}} = 5$$

Oefening 8 :

Bereken die limiete van die volgende funksies :

a.

$$\lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{x - 1} \right) =$$

b.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x} =$$

Voorbeelde :

1. Gegee : $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

By $x = 2$:

$$\frac{x^2 - 4}{x - 2} = \frac{2^2 - 4}{2 - 2} = \frac{0}{0} \text{ (ongedefineerd)}$$

By $x = 2,1$: $\frac{(2,1)^2 - 4}{2,1 - 2} = 4,100$

By $x = 2,001$: $\frac{(2,001)^2 - 4}{2,001 - 2} = 4,001$

Dus:

$$\lim_{x \rightarrow 2} \frac{x^2 - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2}$$

$$= \lim_{x \rightarrow 2} (x + 2)$$

$$= 4 \Rightarrow$$

Exercise 8 :

Determine the limits of the following functions :

a.

$$\lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{x - 1} \right) =$$

b.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x} =$$

Example :

1. Given : $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

At $x = 2$: $\frac{x^2 - 4}{x - 2} = \frac{2^2 - 4}{2 - 2} = \frac{0}{0}$ (undefined)

At $x = 2,1$: $\frac{(2,1)^2 - 4}{2,1 - 2} = 4,100$

At $x = 2,001$: $\frac{(2,001)^2 - 4}{2,001 - 2} = 4,001$

Thus:

$$\lim_{x \rightarrow 2} \frac{x^2 - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2}$$

$$= \lim_{x \rightarrow 2} (x + 2)$$

$$= 4 \Rightarrow$$

2.

$$\begin{aligned}
 & \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (2x + h) \\
 &= 2x
 \end{aligned}$$

3. As

$$y(x) = 3x^2 - 2x + 6, \text{ is}$$

$$\begin{aligned}
 y(x+h) &= 3(x+h)^2 - 2(x+h) + 6 \\
 &= 3(x^2 + 2xh + h^2) - 2x - 2h + 6 \\
 &= 3x^2 + 6xh + 3h^2 - 2x - 2h + 6
 \end{aligned}$$

$$\text{Bepaal } \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h}$$

2.

$$\begin{aligned}
 & \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (2x + h) \\
 &= 2x
 \end{aligned}$$

3. If

$$y(x) = 3x^2 - 2x + 6, \text{ is}$$

$$\begin{aligned}
 y(x+h) &= 3(x+h)^2 - 2(x+h) + 6 \\
 &= 3(x^2 + 2xh + h^2) - 2x - 2h + 6 \\
 &= 3x^2 + 6xh + 3h^2 - 2x - 2h + 6
 \end{aligned}$$

$$\text{Determine } \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h}$$

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 2(x+h) + 6] - [3x^2 - 2x + 6]}{h} \\
&= \lim_{h \rightarrow 0} \frac{[3(x^2 + 2xh + h^2) - 2(x+h) + 6] - [3x^2 - 2x + 6]}{h} \\
&= \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 - 2x - 2h + 6) - 3x^2 + 2x - 6}{h} \\
&= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h} \\
&= \lim_{h \rightarrow 0} (6x + 3h - 2) \\
&= 6x - 2
\end{aligned}$$

Oefening 9 :

i.

As $y(x) = x^2 - 3x$, bepaal $\lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h}$

ii.

As $f(x) = 2x + 7$ en $a = 3$,

bepaal $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$

3.2 Helling van 'n Reguitlyn

Die vergelyking vir 'n reguitlyn is $y = mx + c$, dit wil sê y is 'n funksie van x , of $y = f(x)$.

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 2(x+h) + 6] - [3x^2 - 2x + 6]}{h} \\
&= \lim_{h \rightarrow 0} \frac{[3(x^2 + 2xh + h^2) - 2(x+h) + 6] - [3x^2 - 2x + 6]}{h} \\
&= \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 - 2x - 2h + 6) - 3x^2 + 2x - 6}{h} \\
&= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h} \\
&= \lim_{h \rightarrow 0} (6x + 3h - 2) \\
&= 6x - 2
\end{aligned}$$

Exercise 9 :

i.

If $y(x) = x^2 - 3x$, determine $\lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h}$

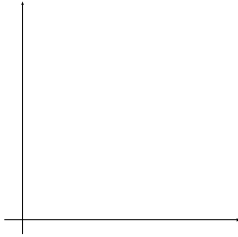
ii.

If $f(x) = 2x + 7$ and $a = 3$,

determine $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$

3.2 Gradient of a straight line

The equation for a straight line is $y = mx + c$, in other words y is a function of x , or $y = f(x)$.



Beskou punt A by $x=x_0$, dan is $y = y_1$
 en punt B by $x=x+h$, dan is $y = y_2$

As $y = f(x)$
 is $y_1 = f(x_0)$
 en $y_2 = f(x_0 + h)$
 waar $x_2 - x_1 = \Delta x = h$

$$\text{Helling} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

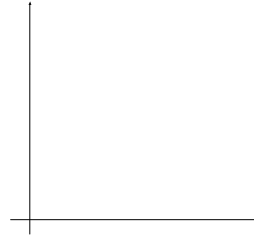
$$\therefore \text{Helling} = \frac{f(x_0 + h) - f(x_0)}{h}$$

Voorbeeld :

Vir 'n reguitlyn is $y = f(x)$

$$= mx + c$$

$$\begin{aligned} \therefore \text{Helling} &= \frac{[m(x_0 + h) + c] - [m(x_0) + c]}{h} \\ &= \frac{mx_0 + mh + c - mx_0 - c}{h} \\ &= \frac{mh}{h} \\ &= m \Rightarrow \end{aligned}$$



Consider point A by $x=x_0$, then $y = y_1$
 and point B at $x=x+h$, then $y = y_2$

If $y = f(x)$
 then $y_1 = f(x_0)$
 and $y_2 = f(x_0 + h)$
 where $x_2 - x_1 = \Delta x = h$

$$\text{Gradient} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\therefore \text{Gradient} = \frac{f(x_0 + h) - f(x_0)}{h}$$

Example :

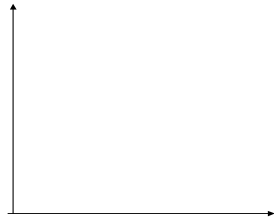
For a straight line is $y = f(x)$

$$= mx + c$$

$$\begin{aligned} \therefore \text{Gradient} &= \frac{[m(x_0 + h) + c] - [m(x_0) + c]}{h} \\ &= \frac{mx_0 + mh + c - mx_0 - c}{h} \\ &= \frac{mh}{h} \\ &= m \Rightarrow \end{aligned}$$

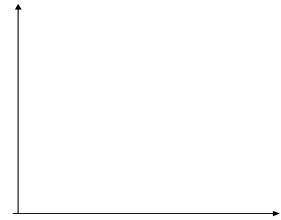
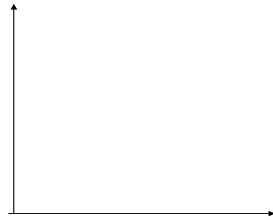
Opermerking :

$$\text{Helling} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_2 > x_1$$



Remark:

$$\text{Gradient} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_2 > x_1$$



3.3 Helling van 'n kromme

Beskou die vergelyking $y = f(x)$, waar

$$y = x^2 - 3x + 2$$

	Q	Q'	Q''	Q'''	Q''''	Q'''''	P	S'	S''	S'''
x = 6	5	4	3	2,6	2,2	2	1	0	-1	
y = 20	12	6	2	1,6	1,2	0	0	2	7	

Teken die Grafiek

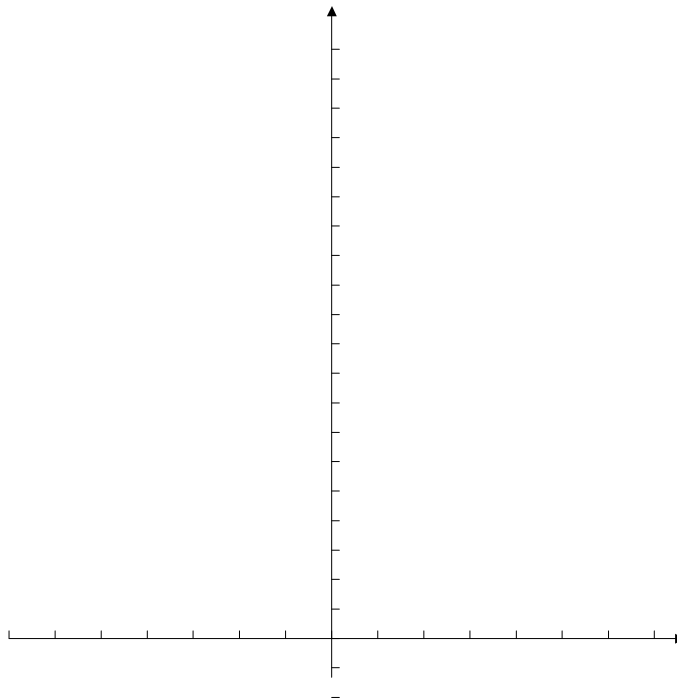
3.3 Gradient of a curved

Consider the equation $y = f(x)$, where:

$$y = x^2 - 3x + 2$$

	Q	Q'	Q''	Q'''	Q''''	Q'''''	P	S'	S''	S'''
x = 6	5	4	3	2,6	2,2	2	1	0	-1	
y = 20	12	6	2	1,6	1,2	0	0	2	7	

Draw the Graph



Laat Q al nader na P toe beweeg. Verbind die punte PQ, PQ¹, ens. en bepaal die hellings.

$$M_{PQ} =$$

$$M_{PQ^1} =$$

$$M_{PQ''} =$$

$$M_{PQ'''} =$$

$$M_{PQ''''} =$$

$$M_{PQ'''''} =$$

Wanneer P en Q saamval, is

$$m = \frac{\Delta y}{\Delta x} = \frac{0}{0} \quad (\text{ongedefineerd})$$

Gevolgtrekking:

Soos wat $\Delta x \rightarrow 0$ word die helling nader aan 1 óf as $\Delta x \rightarrow 0$, dan strewe die helling na 'n limiet waarde.

$$\therefore \text{Helling} = m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

Om dus die helling by punt P te bepaal, dit wil sê in die limiet wanneer $\Delta x \rightarrow 0$, word die **helling van die raaklyn** aan die kromme bepaal.

Die helling word dus as volg gedefinieer:

$$\text{helling} = m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ \left(= \Delta x \rightarrow 0 \frac{y_2 - y_1}{x_2 - x_1} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \right)$$

Let Q move closer to P. Connect the dots PQ, PQ¹, etc. and determine the gradients.

$$M_{PQ} =$$

$$M_{PQ^1} =$$

$$M_{PQ''} =$$

$$M_{PQ'''} =$$

$$M_{PQ''''} =$$

$$M_{PQ'''''} =$$

When P and Q coincide, then

$$m = \frac{\Delta y}{\Delta x} = \frac{0}{0} \quad (\text{undefined})$$

Conclusion :

When $\Delta x \rightarrow 0$ the gradient is approaching the value of 1, ór when $\Delta x \rightarrow 0$ then the gradient is approaching a limit value.

$$\therefore \text{Gradient} = m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

In order to determine the gradient at point P, i.e. in the limit when $\Delta x \rightarrow 0$, then we determine the **gradient of the curve** at that point.

The definition of a gradient is as follows:

$$\text{gradient} = m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ \left(= \Delta x \rightarrow 0 \frac{y_2 - y_1}{x_2 - x_1} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \right)$$

Voorbeeld :

$$y = f(x) = x^2 - 3x + 2$$

Bepaal die helling van die raaklyn by punt

S (0,2) en Q(3,2)

$$\begin{aligned} M &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - (x^2 - 3x + 2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3x - 3h + 2 - x^2 + 3x - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 + 2xh - 3h}{h} \\ &= \lim_{h \rightarrow 0} h + 2x - 3 \end{aligned}$$

$m = 2x - 3$,
sodat by punt S (0,2) is die helling van die raaklyn aan die kromme:

$$\begin{aligned} m &= 2(0) - 3 \\ &= 0 - 3 \\ &= -3 \Rightarrow \end{aligned}$$

en by punt Q''' is die helling

$$\begin{aligned} m &= 2(3) - 3 \\ &= 6 - 3 \\ &= +3 \Rightarrow \end{aligned}$$

Opmerking

Die helling van die raaklyn aan die kromme het by verskillende punte op die kromme verskillende waardes. Die helling is dus 'n funksie van x , soos blyk uit vgl (3.1).

So 'n funksie (soos 3.1) wat die helling uitdruk in terme van x , word 'n **hellingsfunksie** of **afgeleide funksie** of **afgeleide** genoem en word geskryf as:

$$\frac{d}{dx}(y), \text{ of } \frac{dy}{dx} \text{ of } \dot{y} \text{ of } f'(x)$$

Example :

$$y = f(x) = x^2 - 3x + 2$$

Determine the gradient of the tangent at point

S (0,2) en Q(3,2)

$$\begin{aligned} M &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - (x^2 - 3x + 2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3x - 3h + 2 - x^2 + 3x - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 + 2xh - 3h}{h} \\ &= \lim_{h \rightarrow 0} h + 2x - 3 \end{aligned}$$

$m = 2x - 3$,
therefore at point S (0,2) is the gradient of the tangent at the curve:

$$\begin{aligned} m &= 2(0) - 3 \\ &= 0 - 3 \\ &= -3 \Rightarrow \end{aligned}$$

and at point Q''' is the gradient

$$\begin{aligned} m &= 2(3) - 3 \\ &= 6 - 3 \\ &= +3 \Rightarrow \end{aligned}$$

Remark :

The gradient of the tangent to the curve has different values at different points on the curve. Therefore the gradient is a function of x , as is shown in equation (3.1).

A function (such as 3.1) which gives the gradient in terms of x , is called a **gradient function** or a **derived function** and can be written as:

$$\frac{d}{dx}(y), \text{ of } \frac{dy}{dx} \text{ of } \dot{y} \text{ of } f'(x)$$

In bostaande voorbeeld is die vergelyking :

$$y = x^2 - 3x + 2 \text{ en die afgeleide :}$$

$$\frac{dy}{dx} = 2x - 3$$

Opermerking:

$$\text{Met } \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

kan aangetoon word dat:

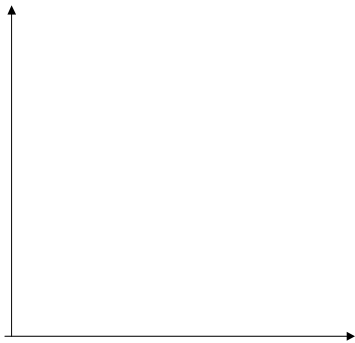
$$\text{As } y = x^2,$$

$$\text{As } y = 3x$$

$$\text{is } \frac{dy}{dx} = 2x,$$

$$\text{is } \frac{dy}{dx} = 3$$

$$\text{As } y = \text{konstante} = k \text{ is } \frac{dy}{dx} = 0$$



Bewys :

$$\text{As } y = f(x) = k$$

$$\text{dan is } f(x+h) = k$$

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{k - k}{h} \\ \text{sodat} \quad &= \lim_{h \rightarrow 0} 0 \\ &= 0 \Rightarrow \end{aligned}$$

In above example the equation is :

$$y = x^2 - 3x + 2 \text{ en die afgeleide :}$$

$$\frac{dy}{dx} = 2x - 3$$

Remark :

$$\text{With } \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

it can be shown that:

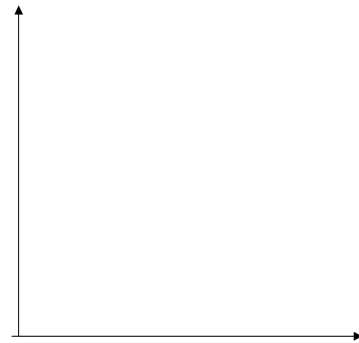
$$\text{If } y = x^2,$$

$$\text{If } y = 3x$$

$$\text{then } \frac{dy}{dx} = 2x,$$

$$\text{is } \frac{dy}{dx} = 3$$

$$\text{When } y = \text{constant} = k \text{ then } \frac{dy}{dx} = 0$$



Proof :

$$\text{If } y = f(x) = k$$

$$\text{then } f(x+h) = k$$

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{k - k}{h} \\ \text{so that} \quad &= \lim_{h \rightarrow 0} 0 \\ &= 0 \Rightarrow \end{aligned}$$

3.4 Afgeleides van Bekende Funksies

Inspeksie van die voorgaande voorbeeld lei dus tot die gevolgtrekking dat vir:

$$y = x^n \quad , \text{ is } \frac{dy}{dx} = nx^{n-1} \quad (3.4.1)$$

$$y = ax^n \quad , \text{ is } \frac{dy}{dx} = anx^{n-1} \quad (3.4.2)$$

Op dieselfde wyse kan verder aangetoon word dat vir:

$$y = \sin x \quad , \text{ is } \frac{dy}{dx} = \cos x \quad (3.4.3)$$

$$y = \sin ax \quad , \text{ is } \frac{dy}{dx} = a \cos ax \quad (3.4.4)$$

$$y = \cos x \quad , \text{ is } \frac{dy}{dx} = -\sin x \quad (3.4.5)$$

$$y = \cos ax \quad , \text{ is } \frac{dy}{dx} = -a \sin ax \quad (3.4.6)$$

$$y = \ln x \quad , \text{ is } \frac{dy}{dx} = \frac{1}{x} \quad (3.4.7)$$

$$y = a^x \quad , \text{ is } \frac{dy}{dx} = (\ln a)a^x \quad (3.4.8)$$

$$y = e^x \quad , \text{ is } \frac{dy}{dx} = e^x \quad (3.4.9)$$

$$y = a^{f(x)} \quad , \text{ is } \frac{dy}{dx} = f'(x)e^{f(x)} \quad (3.4.10)$$

Voorbeelde :

$$\text{a. } \frac{d}{dx}(x^5) = 5x^{5-1} = 5x^4$$

$$\text{b. } \frac{d}{dx}(5x^5) = 25x^{5-1} = 25x^4$$

$$\text{c. } \frac{d}{dx}(5) = 0$$

$$\text{d. } \frac{d}{dx}\left(\frac{1}{x^5}\right) = \frac{d}{dx}(x^{-5}) = -5x^{-5-1} = -5x^{-6}$$

$$\text{e. } \frac{d}{dx}(\sin 2x) = 2 \cos 2x$$

$$\text{f. } \frac{d}{dx}(\cos 2x) = -2 \sin 2x$$

3.4 Derivatives of known Functions

Inspection of the previous example leads to the conclusion that for:

$$y = x^n \quad , \text{ is } \frac{dy}{dx} = nx^{n-1} \quad (3.4.1)$$

$$y = ax^n \quad , \text{ is } \frac{dy}{dx} = anx^{n-1} \quad (3.4.2)$$

In the same way it can be shown that for:

$$y = \sin x \quad , \text{ is } \frac{dy}{dx} = \cos x \quad (3.4.3)$$

$$y = \sin ax \quad , \text{ is } \frac{dy}{dx} = a \cos ax \quad (3.4.4)$$

$$y = \cos x \quad , \text{ is } \frac{dy}{dx} = -\sin x \quad (3.4.5)$$

$$y = \cos ax \quad , \text{ is } \frac{dy}{dx} = -a \sin ax \quad (3.4.6)$$

$$y = \ln x \quad , \text{ is } \frac{dy}{dx} = \frac{1}{x} \quad (3.4.7)$$

$$y = a^x \quad , \text{ is } \frac{dy}{dx} = (\ln a)a^x \quad (3.4.8)$$

$$y = e^x \quad , \text{ is } \frac{dy}{dx} = e^x \quad (3.4.9)$$

$$y = a^{f(x)} \quad , \text{ is } \frac{dy}{dx} = f'(x)e^{f(x)} \quad (3.4.10)$$

Examples :

$$\text{a. } \frac{d}{dx}(x^5) = 5x^{5-1} = 5x^4$$

$$\text{b. } \frac{d}{dx}(5x^5) = 25x^{5-1} = 25x^4$$

$$\text{c. } \frac{d}{dx}(5) = 0$$

$$\text{d. } \frac{d}{dx}\left(\frac{1}{x^5}\right) = \frac{d}{dx}(x^{-5}) = -5x^{-5-1} = -5x^{-6}$$

$$\text{e. } \frac{d}{dx}(\sin 2x) = 2 \cos 2x$$

$$\text{f. } \frac{d}{dx}(\cos 2x) = -2 \sin 2x$$

Oefeninge :

Differentieer die volgende met betrekking tot t.

a. $y = \frac{1}{t}$

b. $y = \sqrt{t^5}$

c. $y = 2^\pi$

d. $y = e^{t^3}$

f. $y = \cos^2 t - \sin^2 t$ (Sien 2.4)

g. $y = 3t^2 - 4t + 12$

Exercise :

Differentiate the following with regard to t.

a. $y = \frac{1}{t}$

b. $y = \sqrt{t^5}$

c. $y = 2^\pi$

d. $y = e^{t^3}$

f. $y = \cos^2 t - \sin^2 t$ (See 2.4)

g. $y = 3t^2 - 4t + 12$

3.5 Differensiasiereëls3.5.1 **Somreël**

Beskou $f = f(x)$ en $g = g(x)$

Tel die twee funksies bymekaar, sodat hulle som 'n nuwe funksie vorm, naamlik:

$$y(x) = f(x) + g(x)$$

Wanneer ons differensieër is

$$\frac{dy}{dx} = \frac{df}{dx} + \frac{dg}{dx} \quad \text{somreël}$$

Voorbeeld :

$$f(x) = 2x^2 + x + 3$$

$$\frac{df}{dx} = 4x + 1$$

$$g(x) = 3x^2 + 2x + 1$$

$$\frac{dg}{dx} = 6x + 2$$

Dan is (somreël)

$$\begin{aligned} \frac{dy}{dx} &= (4x + 1) + (6x + 2) \\ &= 10x + 3 \Rightarrow \end{aligned}$$

3.5 Differentiation Rule3.5.1 **Sum Rule**

Consider $f = f(x)$ and $g = g(x)$

Add the two functions so that their sum will form a new function, namely:

$$y(x) = f(x) + g(x)$$

When we differentiate

$$\frac{dy}{dx} = \frac{df}{dx} + \frac{dg}{dx} \quad \text{sum rule}$$

Example :

$$f(x) = 2x^2 + x + 3$$

$$\frac{df}{dx} = 4x + 1$$

$$g(x) = 3x^2 + 2x + 1$$

$$\frac{dg}{dx} = 6x + 2$$

Then is (sum rule)

$$\begin{aligned} \frac{dy}{dx} &= (4x + 1) + (6x + 2) \\ &= 10x + 3 \Rightarrow \end{aligned}$$

wat dieselfde is as

$$y = (2x^2 + x + 3) + (3x^2 + 2x + 1) \\ = 5x^2 + 3x + 4$$

sodat

$$\frac{dy}{dx} = \frac{d}{dx}(5x^2 + 3x + 4) \\ = 10x + 3 \Rightarrow$$

Oefening :

a. Met

$$f(x) = 3x^{-2} \quad \text{en} \quad g(x) = -4x^4$$

i. Skryf die som $y(x)$ neer

ii. Bepaal $\frac{dy}{dx}$

b. Bepaal $\frac{d}{dx}(x + \cos x)$

c. Bepaal $\frac{d}{dx}(3 \cos x + \sin 2x)$

d. Bepaal $\frac{d}{dx}(x^3 + 2x - 5)$

3.5.2 Prodrukreël

$$\text{Laat } y(x) = f(x) \cdot g(x)$$

Wanneer ons differensieër is

$$\frac{dy}{dx} = f \frac{dg}{dx} + g \frac{df}{dx} \quad \text{prodrukreël}$$

wich is the same as

$$y = (2x^2 + x + 3) + (3x^2 + 2x + 1) \\ = 5x^2 + 3x + 4$$

so that

$$\frac{dy}{dx} = \frac{d}{dx}(5x^2 + 3x + 4) \\ = 10x + 3 \Rightarrow$$

Exercise :

a. With

$$f(x) = 3x^{-2} \quad \text{and} \quad g(x) = -4x^4$$

i. Write the sum $y(x)$ down

ii. Determine $\frac{dy}{dx}$

b. Determine $\frac{d}{dx}(x + \cos x)$

c. Determine $\frac{d}{dx}(3 \cos x + \sin 2x)$

d. Determine $\frac{d}{dx}(x^3 + 2x - 5)$

3.5.2 Product Rule

$$\text{Let } y(x) = f(x) \cdot g(x)$$

When we differentiate then

$$\frac{dy}{dx} = f \frac{dg}{dx} + g \frac{df}{dx} \quad \text{product rule}$$

Voorbeeld :

$$y = (3x^2 + 5x)(2x^3 - 3)$$

$$\begin{aligned}\frac{dy}{dx} &= (3x^2 + 5x)(6x^2) + (2x^3 - 3)(6x + 5) \\ &= 18x^4 + 30x^3 + (12x^4 + 10x^3 - 18x - 15) \\ &= 18x^4 + 30x^3 + 12x^4 + 10x^3 - 18x - 15 \\ &= 30x^4 + 40x^3 - 18x - 15\end{aligned}$$

Oefening :

a. $\frac{d}{dx}(x^2 \sin x)$

b. $\frac{d}{dx}(2\sqrt{x} \ln x)$

c. $y = \cos^2 x$

3.5.3 Kwosiëntreël

Laat $y(x) = \frac{f(x)}{g(x)}$

Wanneer ons differensieër is

$$\frac{dy}{dx} = \frac{g\left(\frac{df}{dx}\right) - f\left(\frac{dg}{dx}\right)}{g^2} \quad \text{Kwosiëntreël}$$

Voorbeeld :

Laat $y(x) = \frac{9x^7}{(x^2 + 1)}$

dan is $f(x) = 9x^7$, $\frac{df}{dx} = 63x^6$

en $g(x) = (x^2 + 1)$, $\frac{dg}{dx} = 2x$

Example :

$$y = (3x^2 + 5x)(2x^3 - 3)$$

$$\begin{aligned}\frac{dy}{dx} &= (3x^2 + 5x)(6x^2) + (2x^3 - 3)(6x + 5) \\ &= 18x^4 + 30x^3 + (12x^4 + 10x^3 - 18x - 15) \\ &= 18x^4 + 30x^3 + 12x^4 + 10x^3 - 18x - 15 \\ &= 30x^4 + 40x^3 - 18x - 15\end{aligned}$$

Exercise :

a. $\frac{d}{dx}(x^2 \sin x)$

b. $\frac{d}{dx}(2\sqrt{x} \ln x)$

c. $y = \cos^2 x$

3.5.4 Quotient Rule

Let $y(x) = \frac{f(x)}{g(x)}$

When we differentiate then

$$\frac{dy}{dx} = \frac{g\left(\frac{df}{dx}\right) - f\left(\frac{dg}{dx}\right)}{g^2} \quad \text{Quotient rule}$$

Example :

Let $y(x) = \frac{9x^7}{(x^2 + 1)}$

then is $f(x) = 9x^7$, $\frac{df}{dx} = 63x^6$

and $g(x) = (x^2 + 1)$, $\frac{dg}{dx} = 2x$

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x^2+1)63x^6 - 9x^7(2x)}{(x^2+1)^2} \\ \text{sodat} \quad &= \frac{63x^8 + 63x^6 - 18x^8}{(x^2+1)^2} \\ &= \frac{45x^8 + 63x^6}{(x^2+1)^2} \Rightarrow\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x^2+1)63x^6 - 9x^7(2x)}{(x^2+1)^2} \\ \text{so that} \quad &= \frac{63x^8 + 63x^6 - 18x^8}{(x^2+1)^2} \\ &= \frac{45x^8 + 63x^6}{(x^2+1)^2} \Rightarrow\end{aligned}$$

Oefening :

a. Toon aan dat $\frac{d}{dx} (\tan x) = \sec^2 x$

b. $y = \frac{5x-3}{7x+2}$

Bepaal $\frac{dy}{dx}$

3.5.4 **Kettingreël**

Soms is dit nodig om 'n **saamgestelde funksie** te diffirensieër. Met die reeds bekende tegnieke is dit moeilik uitvoerbaar, bv.

$$y = \sin 4x^2$$

Ons kan wel iets soos $\sin x$ en $4x^2$ diffirensieër.

Ons maak dan gebruik van **substitusie**.

Stel $u = 4x^2$, sodat $\frac{du}{dx} = 8x$ en

dan is $y = \sin u$, wat wel maklik

gediffirensieër kan word, sodat $\frac{dy}{du} = \cos u$

Nou is $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ Kettingreël

Exercise :

a. Show that $\frac{d}{dx} (\tan x) = \sec^2 x$

b. $y = \frac{5x-3}{7x+2}$

Determine $\frac{dy}{dx}$

3.5.4 **Chain Rule**

Sometimes it is necessary to differentiate a **combined function**. With the already known techniques it is difficult to implement, e.g.

$$y = \sin 4x^2$$

However, we can differentiate functions like $\sin x$ and $4x^2$.

We then make use of **substitution**.

Set $u = 4x^2$, so that $\frac{du}{dx} = 8x$ en

then $y = \sin u$, which can easily be

differentiated, so that $\frac{dy}{du} = \cos u$

Now is $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ Chain rule

Voorbeeld:

a. In bostaande voorbeeld het ons dan:

$$\begin{aligned}\frac{dy}{dx} &= (\cos u)(8x) \\ &= \cos(4x^2)(8x) \\ &= 8x \cos 4x^2 \Rightarrow\end{aligned}$$

b.

$$y = (x - 2x^3)^{-11}, \quad \text{Bepaal } \frac{dy}{dx}$$

$$\text{Stel } u = x - 2x^3, \quad \frac{du}{dx} = 1 - 6x^2$$

$$\text{dan is } y = u^{-11}, \quad \frac{dy}{du} = -11 u^{-12}$$

$$\begin{aligned}\text{sodat } \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= (-11 u^{-12})(1 - 6x^2) \\ &= -11(x - 2x^3)^{-12}(1 - 6x^2) \\ &= (-11 + 66x^2)(x - 2x^3)^{-12}\end{aligned}$$

c.

$$y = \cos^3 4x$$

$$\text{Stel } v = 4x, \quad \frac{dv}{dx} = 4$$

$$\text{Stel } u = \cos 4x,$$

$$= \cos v, \quad \text{sodat } \frac{du}{dv} = -\sin v$$

$$\text{Nou is } y = u^3, \quad \frac{dy}{du} = 3u^2$$

$$\begin{aligned}\text{Sodat } \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx} \quad (\text{Kettingreël}) \\ &= (3u^2)(-\sin v)(4) \\ &= 3(\cos^2 4x)(-\sin 4x)4 \\ &= -12 \cos^2 4x \sin 4x \Rightarrow\end{aligned}$$

Example :

a. In the example above we then have :

$$\begin{aligned}\frac{dy}{dx} &= (\cos u)(8x) \\ &= \cos(4x^2)(8x) \\ &= 8x \cos 4x^2 \Rightarrow\end{aligned}$$

b.

$$y = (x - 2x^3)^{-11}, \quad \text{Determine } \frac{dy}{dx}$$

$$\text{Stel } u = x - 2x^3, \quad \frac{du}{dx} = 1 - 6x^2$$

$$\text{so that } y = u^{-11}, \quad \frac{dy}{du} = -11 u^{-12}$$

$$\begin{aligned}\text{sothat } \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= (-11 u^{-12})(1 - 6x^2) \\ &= -11(x - 2x^3)^{-12}(1 - 6x^2) \\ &= (-11 + 66x^2)(x - 2x^3)^{-12}\end{aligned}$$

c.

$$y = \cos^3 4x$$

$$\text{Set } v = 4x, \quad \frac{dv}{dx} = 4$$

$$\text{Set } u = \cos 4x,$$

$$= \cos v, \quad \text{so that } \frac{du}{dv} = -\sin v$$

$$\text{Now is } y = u^3, \quad \frac{dy}{du} = 3u^2$$

$$\begin{aligned}\text{So that } \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx} \quad (\text{Chain rule}) \\ &= (3u^2)(-\sin v)(4) \\ &= 3(\cos^2 4x)(-\sin 4x)4 \\ &= -12 \cos^2 4x \sin 4x \Rightarrow\end{aligned}$$

Oefening :

Gebruik die kettingreël en bepaal die afgeleides van die volgende funksies:

- a. $y = \sin 3x$
- b. $y = \cos x^5$
- c. $y = \cos \sqrt{x}$
- d. $y = \ln(-x)$
- e. $y = e^{2x^3}$

3.5.5 Hoër Orde Afgeleides

In die voorbeeld in paragraaf 3.3 was

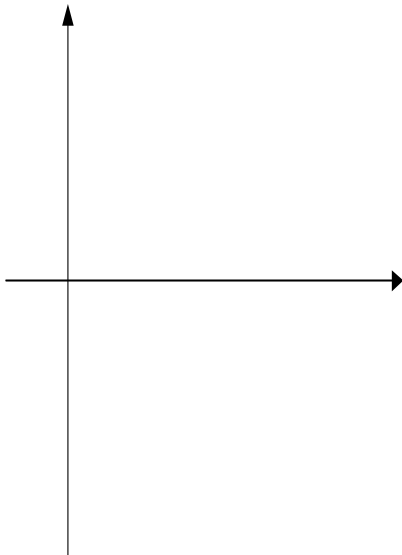
$$y = x^2 - 3x + 2$$

sodat $\frac{dy}{dx} = 2x - 3$

Hierdie is nou 'n nuwe funksie van x , nl. $g(x)$

waar $g(x) = 2x - 3 =$ hellingsfunksie

Teken die grafiek van hierdie funksie:

Exercise :

Use the chain rule and determine the derivatives of the following functions :

- a. $y = \sin 3x$
- b. $y = \cos x^5$
- c. $y = \cos \sqrt{x}$
- d. $y = \ln(-x)$
- e. $y = e^{2x^3}$

3.5.5 Higher Order Derivatives

In the example in paragraph 3.3 was

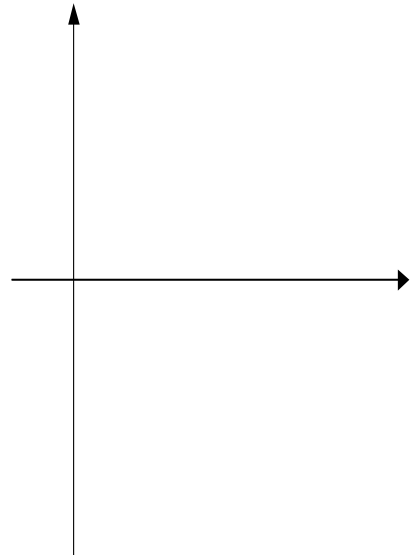
$$y = x^2 - 3x + 2$$

so that $\frac{dy}{dx} = 2x - 3$

This is now a new function of x , $g(x)$

where $g(x) = 2x - 3 =$ gradient function

Draw a graph of this function:



Bepaal die helling $\frac{\Delta y}{\Delta x}$ van hierdie grafiek:

$$m =$$

$$\text{maar } g = 2x - 3$$

$$\text{sodat } \frac{dg}{dx} = 2$$

dit wil sê die **helling** van die **hellingsfunksie** is 2.

Uit bostaande volg dus dat:

$$\frac{d}{dx}(g) = \frac{d}{dx}\left(\frac{dy}{dx}\right) = 2$$

Hierdie word die **tweede afgeleide** van y genoem en word geskryf as:

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}, \frac{d^2}{dx^2}(y), y'', f''$$

Bogenoemde kan natuurlik wéér gedifferensieer word sodat hoër orde afgeleides bepaal kan word.

$$\begin{aligned} \therefore \frac{d}{dx}\left(\frac{dg}{dx}\right) &= \frac{d}{dx}\left(\frac{d}{dx}\left(\frac{dy}{dx}\right)\right) \\ &= \frac{d}{dx}(2) \\ &= 0 \end{aligned}$$

Voorbeeld :

$$y = x^3 + 2x^2 - 3x + 1$$

Vind alle hoër orde afgeleides van y.

$$f(x) = y = x^3 + 2x^2 - 3x + 1$$

$$f^1(x) = 3x^2 + 4x - 3$$

$$f^{11}(x) = 6x + 4$$

$$f^{111}(x) = 6$$

$$f^n(x) = 0 \text{ vir alle } n \geq 4$$

Determine the gradient $\frac{\Delta y}{\Delta x}$ from this graph:

$$m =$$

$$\text{but } g = 2x - 3$$

$$\text{so that } \frac{dg}{dx} = 2$$

in other words the **gradient** of the **gradient function** has the value of 2.

From above it follows that:

$$\frac{d}{dx}(g) = \frac{d}{dx}\left(\frac{dy}{dx}\right) = 2$$

This is called the **second derivative** of y and is written as:

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}, \frac{d^2}{dx^2}(y), y'', f''$$

Abovementioned can be differentiated so that the higher order derivatives can be determined.

$$\begin{aligned} \therefore \frac{d}{dx}\left(\frac{dg}{dx}\right) &= \frac{d}{dx}\left(\frac{d}{dx}\left(\frac{dy}{dx}\right)\right) \\ &= \frac{d}{dx}(2) \\ &= 0 \end{aligned}$$

Example :

$$y = x^3 + 2x^2 - 3x + 1$$

Find all the high order derivatives from y.

$$f(x) = y = x^3 + 2x^2 - 3x + 1$$

$$f^1(x) = 3x^2 + 4x - 3$$

$$f^{11}(x) = 6x + 4$$

$$f^{111}(x) = 6$$

$$f^n(x) = 0 \text{ for all } n \geq 4$$

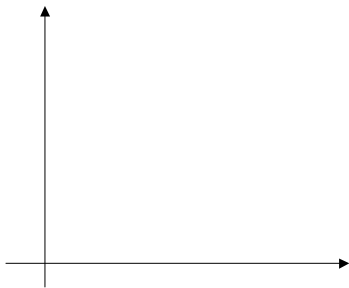
Oefening :

- Bereken $\frac{d^2y}{dx^2}$ as $y = 4x^4 + 2x^2 - 5x - 2$
- Bereken $\frac{d^3y}{dx^3}$ as $y = e^{x^2}$
- Bereken $\frac{d^2y}{dx^2}$ as $y = \frac{x}{x+1}$
- Bereken $\frac{d^3y}{dx^3}$ as $y = 2\sin 3x$
- 'n Reghoeking swembad is 14m lank, 6m breed, 3m diep aan die diep kant en 1 m diep aan die vlak kant. Die swembad word met water gevul teen 'n tempo van $2 \text{ m}^3/\text{min}$. Hoe vinnig styg die watervlak as die water:
 - 2m diep is aan die diep kant? $\left(\frac{1}{21} \text{ m/min}\right)$
 - 2,5 m diep is aan die diep kant? $\left(\frac{1}{42} \text{ m/min}\right)$

3.6 Maksimum en Minimum Waardes

In paragraaf 3.3 het ons die grafiek geteken van die funksie:

$$y = x^2 - 3x + 2$$



Nou is $\frac{dy}{dx} = f'(x) = 2x - 3$

Teken 'n grafiek van f' vir:

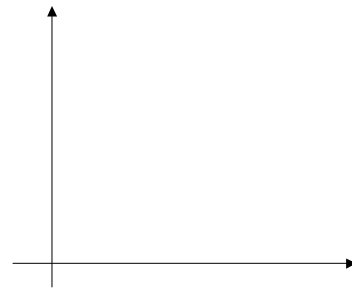
Exercise :

- Determine $\frac{d^2y}{dx^2}$ as $y = 4x^4 + 2x^2 - 5x - 2$
- Determine $\frac{d^3y}{dx^3}$ as $y = e^{x^2}$
- Determine $\frac{d^2y}{dx^2}$ as $y = \frac{x}{x+1}$
- Determine $\frac{d^3y}{dx^3}$ as $y = 2\sin 3x$
- A rectangular swimming pool is 14m in length, 6m in width, 3m deep on the side and 1 m deep on the shallow end. The swimming pool is filled with water at a tempo of $2 \text{ m}^3/\text{min}$. How fast does the water level rise if the water is:
 - 2m deep at the deep end? $\left(\frac{1}{21} \text{ m/min}\right)$
 - 2,5m deep at the deep end? $\left(\frac{1}{42} \text{ m/min}\right)$

3.6 Maximum en Minimum Values

In paragraph 3.3 we drew the graph of the function:

$$y = x^2 - 3x + 2$$



Now is $\frac{dy}{dx} = f'(x) = 2x - 3$

Draw the graph of f' for:

a) $x \geq 2$



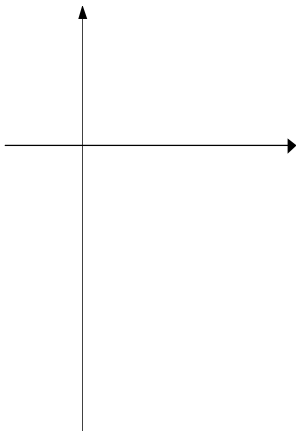
Vir $x \geq 2$ (regs van die draaipunt) is $f^1 > 0$

a) $x \geq 2$



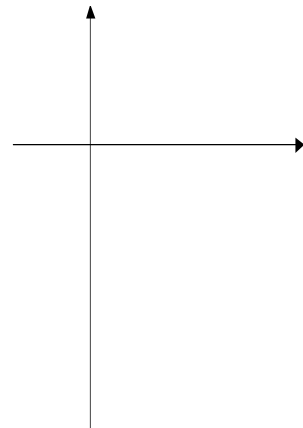
For $x \geq 2$ (right of the turning point) is $f^1 > 0$

b). $x \leq 1$



Vir $x \leq 1$ (links van die draaipunt) is $f^1 < 0$

b). $x \leq 1$



For $x \leq 1$ (left of the turning point) is $f^1 < 0$

Die **draaipunt** van die kromme is waar:

$$\begin{aligned} f^1 &= 0 \\ \therefore 2x - 3 &= 0 \\ 2x &= 3 \\ x &= \frac{3}{2} = 1,5 \end{aligned}$$

Voorbeeld :

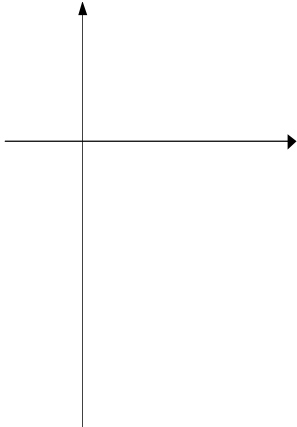
en $y = f(x) = x^3 - 3x - 2$
 $f^1(x) = 3x^2 - 3$

At the turning point of the curved is where:

$$\begin{aligned} f^1 &= 0 \\ \therefore 2x - 3 &= 0 \\ 2x &= 3 \\ x &= \frac{3}{2} = 1,5 \end{aligned}$$

Example :

and $y = f(x) = x^3 - 3x - 2$
 $f^1(x) = 3x^2 - 3$



Deur bloot te kyk na f^1 kan ons nie besluit of ons by 'n **maksimum** of 'n **minimum** is nie.

Bepaal waar die draaipunt(e) van die kromme lê, dit wil sê waar

$$\begin{aligned} f^1 &= 0 \\ \therefore 3x^2 - 3 &= 0 \\ x^2 &= 1 \\ \therefore x &= +1 \quad \text{en} \quad x = -1 \end{aligned}$$

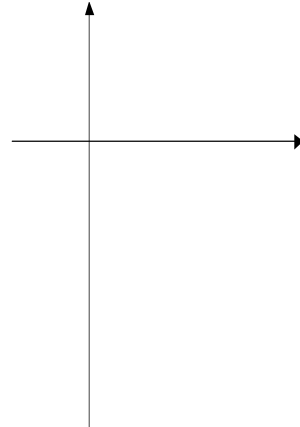
Om te bepaal waar die maksimum en die minimum is, neem:

$$\begin{aligned} f^{11} &= 6x \\ \text{As } x &= -1, \text{ is } f^{11} < 0 \text{ (maksimum)} \\ \text{en as } x &= +1, \text{ is } f^{11} > 0 \text{ (minimum)} \end{aligned}$$

Die teken van die tweede afgeleide bepaal dus of dit 'n maksimum of 'n minimum sal wees.

Voorbeeld :

$$\begin{aligned} y = f(x) &= \frac{1}{4}x^3 - 3x \\ f^1(x) &= \frac{3}{4}x^2 - 3 \\ &= 3\left(\frac{x^2}{4} - 1\right) \\ &= \frac{3}{4}(x^2 - 4) \\ &= \frac{3}{4}(x+2)(x-2) \end{aligned}$$



Just by looking at f^1 we can not decide if we are at a **maximum** or a **minimum**.

Determine where the turning point(s) of the curve lies, in other word where

$$\begin{aligned} f^1 &= 0 \\ \therefore 3x^2 - 3 &= 0 \\ x^2 &= 1 \\ \therefore x &= +1 \quad \text{en} \quad x = -1 \end{aligned}$$

To determine where the maximum and the minimum is, take:

$$\begin{aligned} f^{11} &= 6x \\ \text{If } x &= -1, \text{ is } f^{11} < 0 \text{ (maksimum)} \\ \text{and if } x &= +1, \text{ is } f^{11} > 0 \text{ (minimum)} \end{aligned}$$

The sign of the second derivative determines if it will be a maximum or minimum.

Example :

$$\begin{aligned} y = f(x) &= \frac{1}{4}x^3 - 3x \\ f^1(x) &= \frac{3}{4}x^2 - 3 \\ &= 3\left(\frac{x^2}{4} - 1\right) \\ &= \frac{3}{4}(x^2 - 4) \\ &= \frac{3}{4}(x+2)(x-2) \end{aligned}$$

Die draaipunte lê by $f' = 0$, sodat $(x + 2)(x - 2) = 0$

$$\begin{array}{lcl} \therefore x + 2 = 0 & \text{of} & x - 2 = 0 \\ x = -2 & \text{of} & x = 2 \end{array}$$

Om te bepaal of daar by die draaipunt 'n maksima of minima is, neem

$$\begin{aligned} f'' &= \frac{3}{4}(2x) \\ &= \frac{3}{2}x \end{aligned}$$

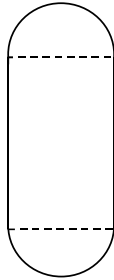
As $x = -2$, is $f'' < 0$ (maksimum)
en as $x = +2$, is $f'' > 0$ (minimum)

Oefening :

- a. Bepaal waar die maksimum en minimum waardes lê vir die funksie

$$f(x) = x^3 - 3x^2 - 24x + 1$$

- b. 'n Sportveld moet ontwerp word sodat dit 'n omtrek van 440 m het. Die vorm van die veld is twee halwe sirkels met 'n reghoekige gedeelte tussenin. Bepaal die afmetings sodat die oppervlakte van die reghoekige gedeelte A 'n maksimum is.



- c. Wanneer iemand hoërs, word die druk P in sy longe hoër. Die vraag ontstaan of sy lugpyp dieselfde bly, groter word of dunner word. Die tempo waarteen lug deur die lugpyp vloei (volume / tyd) word gegee deur:

$$F = k(r_0 - r)r^4, \text{ met } k = \text{konstante}$$

The turn point lies at $f' = 0$, so that $(x + 2)(x - 2) = 0$

$$\begin{array}{lcl} \therefore x + 2 = 0 & \text{or} & x - 2 = 0 \\ x = -2 & \text{or} & x = 2 \end{array}$$

To determine if there is maximum or minimum at the turning point, take

$$\begin{aligned} f'' &= \frac{3}{4}(2x) \\ &= \frac{3}{2}x \end{aligned}$$

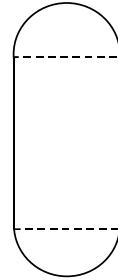
If $x = -2$, is $f'' < 0$ (maximum)
and if $x = +2$, is $f'' > 0$ (minimum)

Exercise :

- a. Determine where the maximum and minimum values lie for the function:

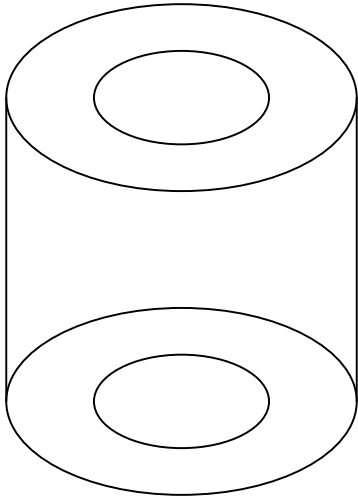
$$f(x) = x^3 - 3x^2 - 24x + 1$$

- b. A sports field must be designed so that the circumference is 440 m. The shape of the field are two half circles with a rectangular portion in between. Determine the measurements so that the area of the rectangular portion A is a maximum.

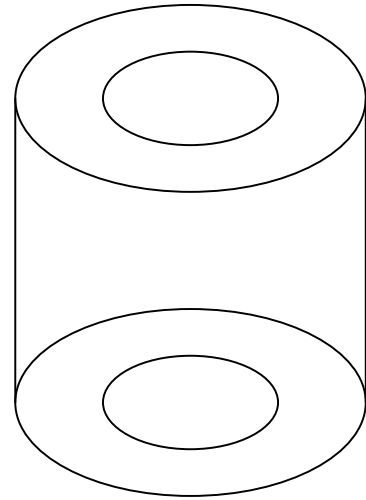


- c. When somebody coughs the pressure in his lungs will increase. The question is if the trachea will remain the same, become bigger or will become thinner. The rate of air flow through the trachea (volume / time) is given by:

$$F = k(r_0 - r)r^4, \text{ with } k = \text{constant}$$



Bepaal die waarde van r by die draaipunt en toon dan aan of die vloeitempo 'n minimum of 'n maksimum is.



Determine the value of r at the turning point and also show if the rate of flow is a minimum or a maximum.